

AI for Education Initiative

Building Local AI Infrastructure for Teachers and Students

Prepared by William Morris | May 2026

Executive Summary

We propose establishing a dedicated AI education server at NAS Jiaxing — a local, privacy-focused platform that gives teachers and students direct access to cutting-edge AI tools without relying on external cloud services or VPNs.

The Problem We Are Solving

Teachers want to use AI in classrooms, but face three barriers:

- Access blocked** — Many AI tools (ChatGPT, Claude, Gemini) are blocked on school networks without VPN
- Data privacy** — Uploading student work to foreign cloud services raises compliance concerns
- Cost and complexity** — Premium AI subscriptions are expensive; teachers don't know where to start

Our Solution

A local AI server running inside the school network, hosting open-source AI models and educational tools that are:

- Always accessible** — No VPN needed; works on school WiFi
- Fully private** — All data stays inside the school; no external cloud uploads
- Free to use** — Open-source software; no subscription fees
- Educational** — Designed specifically for teaching, not general chat

What Is Already Working (Demo)

We have built a proof-of-concept running on a personal workstation. This demonstrates that the technology works and the concept is viable.

Current Demo Machine Specs

Component	Specification
CPU	AMD Ryzen 9 7950X3D (16 cores, 32 threads)

RAM	96GB DDR5
GPU	AMD RX 7900 XTX (24GB VRAM)
Storage	2TB NVMe SSD + 4TB HDD
OS	Ubuntu 24.04 LTS
Network	School LAN — accessible from any classroom
Total Build Cost	~¥20,000 (personal funds)

This machine currently serves educational content and AI tools to the school network. It proves the concept. For a permanent school solution, we recommend a purpose-built server at roughly half the cost.

Proposed School Server — Three Used Enterprise Options

All three options use professional server-grade hardware sourced from Taobao's used enterprise market. All are significantly more cost-effective than the ¥20,000 demo build while delivering far more capability than any new desktop PC at the same price.

★ **Recommended: Option A1 — Rackmount Server**

Best for schools with a server room or rack cabinet. Enterprise-grade 2U chassis with redundant power, 10GbE networking, and maximum expansion.

Option A1: GIGABYTE G292-Z20 Rackmount (~¥12,050) ★

2U enterprise server. Sourced from Taobao used/server refurb market. Tested before shipping. 1-year seller warranty.

Component	Specification	Est. Price
Chassis	GIGABYTE G292-Z20 2U Rackmount Server (used)	¥4,700
Motherboard	MZ22-G20 (SP3 socket, 8x DDR4, 10x PCIe x16)	
CPU	AMD EPYC 7532 — 32 cores / 64 threads, 2.4GHz, 256MB L3 cache	
RAM	256GB DDR4 ECC RDIMM (8x 32GB, used, Samsung/SK Hynix) — expandable to 512GB	¥2,400
GPU	2x NVIDIA Tesla A2 (16GB GDDR6 each, 60W, datacenter AI inference) — expandable to 4x or more	¥4,000
Storage (System)	1TB NVMe SSD (used enterprise, e.g., Intel P4510 / Samsung PM981)	¥400
Storage (Data)	4TB HDD (used enterprise, e.g., HGST / Seagate Exos)	¥350
Power	2200W Dual Redundant PSU (hot-swap, included with chassis)	Included
Cooling	4x Server-grade fans with thermal management (included)	Included
Network	Dual 10GbE SFP+ ports (onboard) + IPMI remote management	Included
Rails / Rack Kit	Sliding rail kit for server cabinet (recommended)	¥200
OS & Software	Ubuntu 24.04 LTS (free, open-source)	¥0
Total	32 cores · 256GB RAM · 32GB VRAM (2x Tesla A2) · Rackmount · 10GbE · Redundant PSU	~¥12,050

Option A2: Supermicro Full-Tower Workstation (~¥13,650)

Dual-socket full-tower server. Best if no server room available — sits on floor or desk like a large PC. 2x the CPU cores of A1.

Component	Specification	Est. Price
Chassis	Supermicro H11-DSI Full-Tower E-ATX (宙斯盾 case)	¥6,800
Motherboard	H11-DSI (dual SP3 socket, ~8x DDR4, ~7x PCIe x16)	
CPU	2x AMD EPYC 7532 — 64 cores / 128 threads total, 2.4GHz	
RAM	256GB DDR4 ECC RDIMM (8x 32GB, used) — expandable to ~512GB	¥2,200
GPU	2x NVIDIA Tesla A2 (16GB GDDR6 each, 60W, datacenter AI inference)	¥4,000
Storage (System)	1TB NVMe SSD (used enterprise, e.g., Samsung PM981)	¥400
Storage (Data)	4TB HDD (used enterprise)	¥350
Power	850W 80Plus Gold single PSU (Great Wall / Huntkey)	Included
Cooling	2x 金钱豹M97 dual-fan tower coolers (SP3)	Included
Network	Dual 1GbE ports (onboard) + IPMI remote management	Included
Rails / Rack Kit	Not applicable (full-tower)	¥0
OS & Software	Ubuntu 24.04 LTS (free, open-source)	¥0
Total	64 cores · 256GB RAM · 32GB VRAM (2x Tesla A2) · Full-Tower · Dual 1GbE	~¥13,650

Option A3: High-Core Full-Tower Workstation (~¥18,850)

Maximum core count. Same chassis as A2 but with 128-core CPUs. Overkill for most school AI workloads — included for comparison.

Component	Specification	Est. Price
Chassis	同泰怡T1DEEP Full-Tower E-ATX (宙斯盾 case)	¥12,000
Motherboard	T1DEEP (dual SP3 socket, 16x DDR4, ~7x PCIe x16)	
CPU	2x AMD EPYC 7B12 — 128 cores / 256 threads total, 2.25GHz (OEM/cloud variant)	
RAM	256GB DDR4 ECC RDIMM (8x 32GB, used) — expandable to 1TB+ via 16 slots	¥2,200
GPU	2x NVIDIA Tesla A2 (16GB GDDR6 each, 60W, datacenter AI inference)	¥4,000
Storage (System)	1TB NVMe SSD (used enterprise)	¥400

Storage (Data)	4TB HDD (used enterprise)	¥350
Power	1250W 80Plus Gold single PSU (Great Wall full-modular)	Included
Cooling	2x 金钱豹M97 dual-fan tower coolers (SP3)	Included
Network	Dual 1GbE ports (onboard) + IPMI remote management	Included
Rails / Rack Kit	Not applicable (full-tower)	¥0
OS & Software	Ubuntu 24.04 LTS (free, open-source)	¥0
Total	128 cores · 256GB RAM · 32GB VRAM (2x Tesla A2) · Full-Tower · Dual 1GbE	~¥18,850

Three Budget Tiers

Each tier uses the same professional server hardware. The difference is configuration scale — choose what fits the school's needs and resources.

Tier	Configuration	Price	Best For
● Entry	A1 Rackmount — 32 cores, 128GB RAM , 1x Tesla A2 (16GB VRAM), 1TB NVMe + 4TB HDD, redundant PSU, 10GbE	~¥8,850	Pilot program, 5–10 teachers, proof of concept
★ Standard	A1 Rackmount — 32 cores, 256GB RAM , 2x Tesla A2 (32GB VRAM), 1TB NVMe + 4TB HDD, redundant PSU, 10GbE	~¥12,050	Full deployment, all teachers and students, long-term growth
Performance	A2 Full-Tower — 64 cores , 256GB RAM, 2x Tesla A2 (32GB VRAM), 1TB NVMe + 4TB HDD, 850W PSU, dual 1GbE	~¥13,650	Maximum concurrent users, no server room available, future CPU-heavy workloads

Upgrade Path

Side-by-Side Comparison

Feature	Option A1: Rackmount ★	Option A2: Full-Tower 64c	Option A3: Full-Tower 128c
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CPU Cores	32 cores / 64 threads	● 64 cores / 128 threads	● 128 cores / 256 threads
Form Factor	● 2U Rackmount (server room)	Full-tower (desk/floor)	Full-tower (desk/floor)
Power Supply	● Dual 2200W redundant (hot-swap)	850W single	1250W single
GPU / VRAM	2x Tesla A2 (32GB total)	2x Tesla A2 (32GB total)	2x Tesla A2 (32GB total)
GPU Expansion	● 10x PCIe slots	~7x PCIe slots	~7x PCIe slots
RAM Slots	8x DDR4 (max 512GB)	~8x DDR4 (max ~512GB)	● 16x DDR4 (max 1TB+)
Drive Bays	● 8x hot-swap 2.5" + NVMe	Limited (internal only)	Limited (internal only)
Network	● Dual 10GbE SFP+	Dual 1GbE	Dual 1GbE
Remote Management	● IPMI/iKVM	IPMI (Supermicro)	IPMI (T1DEEP)
Warranty	1 year (Taobao seller)	1 year (Taobao seller)	1 year (Taobao seller)
CPU Type	Retail EPYC 7532	Retail EPYC 7532	OEM EPYC 7B12 (cloud variant)
Total Price	~¥12,050	~¥13,650	~¥18,850

Why We Recommend the Standard Tier (A1 Rackmount, ~¥12,050)

For a school environment with a server room, the Standard tier is the best long-term investment:

- **Enterprise form factor** — 2U rackmount sits in the server room, not on a desk. Professional, secure, out of the way.
- **Redundant power** — dual 2200W hot-swap PSUs. If one fails, the other keeps running. Zero downtime for students.
- **10GbE networking** — dual SFP+ ports onboard. Future-proof when the school upgrades network infrastructure. 10x faster than 1GbE for large AI model transfers.
- **Maximum expansion** — 10 PCIe slots (can add 2–4 more Tesla A2s later), 8 hot-swap drive bays (add storage without opening the case).
- **Enterprise reliability** — built for 24/7 datacenter operation. Full-tower cases with tower coolers are not designed for sustained 100% load.
- **32 cores is enough** — AI inference is GPU-bound, not CPU-bound. The Tesla A2s will be the bottleneck long before 32 EPYC cores max out.
- **Clear upgrade path** — Start with Entry (~¥8,850) for a pilot, then add RAM and a second GPU to reach Standard (~¥12,050) when proven.

When to choose Entry tier: If the school wants to start small and validate the concept before full

deployment. Entry gives the same rackmount chassis, redundancy, and 10GbE — just less RAM and one GPU. Upgrade later without replacing hardware.

When to choose Performance tier (A2, ~¥13,650): If the school has no server room or rack cabinet. The full-tower A2 delivers 64 cores for only ¥1,600 more than Standard, but loses rackmount, redundant PSU, and 10GbE. Best for offices or classrooms where rackmount is impossible.

A3 is not recommended — 128 cores at ~¥18,850 is overkill for school AI workloads. The ¥6,800 premium over Standard buys CPU cores that will sit idle. AI inference runs on GPUs, not CPUs.

What the Server Will Host

For Teachers — Professional Development

- **AI Lesson Plan Assistant** — Generate differentiated activities, rubrics, and assessments
- **Text Simplifier** — Adapt reading passages for ESL students by level
- **Discussion Starter Generator** — Create engaging prompts for any topic
- **Vocabulary Builder** — Auto-generate word lists with definitions and example sentences
- **Reading Assessment Creator** — Build comprehension questions with answer keys

For Students — Learning Tools

- **Educational Games** — Grammar, vocabulary, and critical thinking games
- **Creative Project Generator** — AI-assisted story writing and project ideas
- **AI Tutor** — Students can ask questions and get explanations without leaving the school network
- **Listening Practice** — AI-generated listening exercises with transcripts

For the School — Infrastructure

- **Internal AI Resource Website** — ai.nasjiaxing.cn — a professional portal for all AI tools
- **Content Management** — IT controls what content is available; everything is curated for education
- **No External Dependencies** — Works even if the school's internet connection is slow or filtered

Benefits to NAS Jiaxing

For Teachers

- Save hours on lesson planning and material creation
- Differentiate instruction automatically
- Access PD workshop materials anytime
- No need for personal VPNs or subscriptions
- Data stays inside the school — fully compliant

For Students

- Personalized learning tools available 24/7 on campus
- Interactive games that reinforce curriculum
- AI writing and creativity support
- Safe environment — no ads, no tracking, no external data sharing

For the School

- Position NAS Jiaxing as a leader in AI education
- One-time hardware cost vs. ongoing subscription fees
- Full control over content and access
- Scalable — add more capabilities as needed

For IT & Compliance

- All traffic stays on the school LAN
- No student data leaves the building
- Can be placed under school's existing ICP备案
- Centrally managed and monitored

Implementation Timeline

Phase	Duration	Actions
Phase 1: Approval & Procurement	2–3 weeks	Principal and IT approve budget; order server hardware
Phase 2: Setup & Configuration	1 week	Install Ubuntu, configure AI software, set up internal website
Phase 3: Content & Testing	1–2 weeks	Load educational tools, test with pilot teachers, gather feedback
Phase 4: Launch & Training	1 week	Internal launch, PD workshop for staff, onboarding materials
Phase 5: Expansion	Ongoing	Add new tools based on teacher requests, student feedback

Compliance & Security

We are committed to full compliance with Chinese regulations:

- **ICP备案** — The site will operate under the school's existing ICP filing as a subdomain (ai.nasjiaxing.cn)
- **Data Sovereignty** — All data stored locally; nothing sent to foreign servers
- **Content Control** — IT and school leadership review and approve all content before go-live
- **No VPN Required** — Everything runs on the school internal network
- **Firewall Friendly** — Standard HTTP on port 80; no special ports or protocols needed

What We Need From the School

Three Requests to Move Forward

1. Budget Approval (~¥10,000)

For a dedicated school AI server. We can source used enterprise hardware to stay within budget.

2. IT Support — Static IP & Subdomain

A reserved static IP on the school LAN for the server, and a DNS entry for ai.nasjiaxing.cn pointing to that IP.

3. ICP Subdomain Authorization

Permission to add ai.nasjiaxing.cn under the school's existing ICP备案.

About the Demo

A live demo is already running on the school network. You can access it from any device connected to the school WiFi at:

`http://10.39.26.217/`

Accessible only from the school LAN

This demo shows the landing page, educational games, AI tools, and the general look and feel of what the final site would be.

About Me — William Morris

I am an English teacher at NAS Jiaxing with a background in technology and a personal interest in AI. I built the demo machine in my own time using personal funds to prove that this is feasible and valuable for our school. I am not asking the school to fund my personal project — I am proposing a separate, school-owned server that serves the entire faculty and student body.

I am happy to manage the technical setup, maintain the server, and train teachers on how to use the tools. My goal is to make NAS Jiaxing a model school for AI-integrated education.

Questions? Contact William Morris at william.morris@nasjiaxing.cn

Full technical details available upon request · Demo accessible on school network